# A CONSTRUCTIVE PROOF OF ZARISKI'S MAIN THEOREM BY FINITE SECTION LIFTING, AND ITS FORMALISATION IN UNIT

Abstract. We prove the algebraic form of Zariski's Main Theorem used in the local structure theory of étale algebras: if $E = R[x_1, \dots, x_n]$ is a finitely generated algebra over a commutative ring $R$, $I$ is an ideal of $R$, and $E/IE$ is finite over $R/I$, then there is $s \in 1 + IE$ such that $R[s, sx_1, \dots, sx_n]$ is finite over $R$ and has the same localisation at $s$ as $E$. The proof is constructive and uses no hypothesis on $R$. It replaces the theorem on formal functions by an explicit finite lifting argument in a Čech complex, driven by constructive noetherianity of a polynomial ring over $\mathbb{Z}$. The last section describes the complete machine-checked proof in Unit.

## 1. The theorem

All rings are commutative. "Finite" means finitely generated as a module. "There is" means mere existence: the proof produces the objects, but it does not claim that they are determined by the data.

**Theorem 1.1.** *Let $R$ be a ring, $I \subseteq R$ an ideal, and $E$ an $R$-algebra generated by $x_1, \dots, x_n$. Suppose that $E/IE$ is finite as an $R/I$-module. Then there is $s \in E$ such that*

(1) *$s - 1 \in IE$;*
(2) *$B = R[s, sx_1, \dots, sx_n] \subseteq E$ is a finite $R$-module;*
(3) *the map $B[1/s] \to E[1/s]$ induced by the inclusion is an isomorphism.*

No noetherian, coherence, field or dimension hypothesis is imposed on $R$, and $E$ need not be finitely presented. With $I = \mathfrak{m}$ for a residually discrete local ring, every prime of $E$ above $\mathfrak{m}$ avoids $s$, which is the form used in the local structure theorem for étale algebras.

**Strategy.** Over a noetherian base, one embeds $\operatorname{Spec} E$ as the chart $Z \neq 0$ of a projective scheme $W$, extends $Z^{-1}$ by zero to a section of $\mathcal{O}_W(-1)$ over each infinitesimal neighbourhood $W \times R/I^k$ of the special fibre (the affine part is open and closed there), and uses the finite approximation consequence of formal functions to find a global section with the same reduction modulo $I$. It need not lift the entire compatible formal section. Clearing denominators in the global section gives the result. We keep the shape of this argument but replace every infinite object by a finite certificate:

- the integrality conclusion follows from a few *degree-lowering relations* (Section 2);
- an arbitrary base reduces to a polynomial ring over $\mathbb{Z}$, which is constructively noetherian (Section 3);
- the sections $Z^{-1}$ modulo $I^k$ are given by explicit formulas (Section 5);
- the formal-functions step becomes one finite dependence relation, produced by a noetherian argument over the Rees algebra (Section 6);
- the corrected section yields the degree-lowering relations (Section 7).

We present the proof backwards from its output.

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## 2. Certificates

Let $A \to F$ be a ring map, $y_1, \dots, y_n \in F$ and $b_1, \dots, b_t \in A$.

**Definition 2.1.** A *ZMT certificate* for $(F, y, b)$ consists of $s \in F$ with $s = 1 + \sum_j b_j v_j$ for some $v_j \in F$, together with monic polynomials $Q_0, Q_1, \dots, Q_n \in A[T]$ such that $Q_0(s) = 0$ and $Q_i(sy_i) = 0$ for every $i$.

**Lemma 2.2.** *Suppose $E$ is generated by $x_1, \dots, x_n$ over $R$ and $(E, x, a)$ has a certificate with all $a_j \in I$. Then $s$ satisfies the conclusion of Theorem 1.1.*

*Proof.* Clearly $s - 1 \in IE$. Put $w_0 = s$, $w_i = sx_i$. Each $w_j$ satisfies the monic equation $Q_j$, of degree $d_j$; hence the monomials $w^e$ with $e_j < d_j$ span $B$ over $R$. The map $B[1/s] \to E[1/s]$ is injective because $B \to E$ is injective. Every $e \in E$ is $H(x)$ for a polynomial $H$ of some degree $N$, and then $s^N H(x)$ is a polynomial in the $w_i$, so $e/s^k = (s^N e)/s^{N+k}$ is in the image. □

**Lemma 2.3** (Determinant trick)**.** *Let $V \subseteq F$ be the $A$-span of a finite family containing $1$. If $u \in F$ satisfies $uV \subseteq V$, then $u$ is a root of a monic polynomial over $A$.*

*Proof.* Write $uv_k = \sum_l \alpha_{lk} v_l$. By the adjugate identity, the characteristic polynomial $\chi$ of $(\alpha_{lk})$ satisfies $\chi(u)v_k = 0$ for all $k$, in particular $\chi(u) = \chi(u)\cdot 1 = 0$. The family need not be independent. □

Let $\mathrm{deh} : A[X_1, \dots, X_n, Z] \to F$ be the ring map $Z \mapsto 1$, $X_i \mapsto y_i$ ("dehomogenisation").

**Proposition 2.4** (Degree lowering)**.** *Suppose $s = 1 + \sum_j b_j v_j$, $s = \mathrm{deh}(\sigma)$ with $\sigma$ homogeneous of degree $D$, and for every $i$*

$$s\, y_i^{a_i+1} = \mathrm{deh}(q_i), \qquad q_i \text{ homogeneous of degree } a_i. \tag{DL}$$

*Then $(F, y, b)$ has a certificate with this $s$.*

*Proof.* Let $d = D + \sum_i a_i$ and let $V$ be the span of the monomials $y^\beta$ with $|\beta| \le d$; it contains $1$. It suffices to show $s\, y^\beta \in V$ whenever $|\beta| \le d + 1$, since then $sV \subseteq V$ and $(sy_i)V \subseteq V$, and Lemma 2.3 applies. Either $\beta_i \ge a_i + 1$ for some $i$, and then $y^\beta = y^\gamma y_i^{a_i+1}$ and $sy^\beta = y^\gamma \mathrm{deh}(q_i)$ has degree at most $|\gamma| + a_i = |\beta| - 1 \le d$; or $|\beta| \le \sum_i a_i$, and then $sy^\beta = \mathrm{deh}(\sigma) y^\beta$ has degree at most $D + \sum_i a_i = d$. This dichotomy is decided on the exponents only. □

Only one finite module is used, for $s$ and all $sy_i$ at once. No ring of global sections and no multiplication table of $B$ is computed.

## 3. Reduction to residual data over $\mathbb{Z}[C]$

**Definition 3.1.** Residual data over a ring $A$ in $n$ variables consist of $b_1, \dots, b_t \in A$, monic $P_1, \dots, P_n \in A[T]$ and polynomials $H_{ij} \in A[Y_1, \dots, Y_n]$. The residual algebra is

$$F(\mathcal{D}) = A[Y_1, \dots, Y_n] \big/ \big(P_i(Y_i) - \textstyle\sum_j b_j H_{ij}(Y)\big)_i,$$

with generators $y_i$, the classes of $Y_i$.

**Lemma 3.2.** *Under the hypotheses of Theorem 1.1 there are residual data over $R$ with all $b_j \in I$ and $P_i(x_i) = \sum_j b_j H_{ij}(x)$ in $E$.*

*Proof.* Lift finitely many generators of $E/IE$ to $g_1, \dots, g_N \in E$ and write $1 \equiv \sum_k \alpha_k g_k$ and $x_i g_k \equiv \sum_l \alpha_{lk}^{(i)} g_l$ modulo $IE$. The characteristic polynomial $P_i$ of $(\alpha_{lk}^{(i)})$ gives $P_i(x_i) g_k \in IE$ for all $k$, hence $P_i(x_i) \in IE$. Collect the finitely many coefficients from $I$ in these memberships into one list $b$, and write the other factors as polynomials in $x$, using that $x$ generates $E$. □

Only finitely many lifts and polynomial expressions are chosen.

**Definition 3.3.** Let $A_0 = \mathbb{Z}[C_1, \ldots, C_M]$ where the $C_k$ stand for all coefficients occurring in residual data $\mathcal{D}$ over $R$: the $b_j$, the non-leading coefficients of the $P_i$ and the coefficients of the $H_{ij}$, each at its own index. Replacing each coefficient by its variable gives universal residual data $\mathcal{D}_0$ over $A_0$ (leading coefficients stay 1), and $h : A_0 \to R$, $C_k \mapsto c_k$, maps $\mathcal{D}_0$ to $\mathcal{D}$.

The residual equations of $\mathcal{D}_0$ hold for $x$ in $E$ through $h$, so there is $\varphi : F(\mathcal{D}_0) \to E$ with $\varphi(y_i) = x_i$. Neither $h$ nor $\varphi$ need be injective, and no kernel is computed.

**Lemma 3.4** (Transport)**.** *A certificate for $(F(\mathcal{D}_0), y, b_0)$ over $A_0$ is sent by $(h, \varphi)$ to a certificate for $(E, x, b)$ over $R$.*

*Proof.* $\varphi(1 + \sum b_{0j} v_j) = 1 + \sum_j b_j \varphi(v_j)$ and $(hQ)(\varphi(u)) = \varphi(Q(u))$; monicity is preserved. □

Recall that a ring is CFN if it is coherent, strongly discrete and constructively noetherian, in the sense of Coquand and Persson. The ring $\mathbb{Z}$ is CFN, and so is a polynomial ring over a CFN ring (Hilbert's basis theorem in this form). Hence $A_0$ is CFN. By Lemmas 2.2, 3.2 and transport, Theorem 1.1 reduces to:

**Theorem 3.5** (Universal ZMT)**.** *For every CFN ring $A$, every $n$ and all residual data $\mathcal{D}$ over $A$ in $n$ variables, $(F(\mathcal{D}), y, b)$ merely has a certificate.*

**Lemma 3.6** (Padding)**.** *Theorem 3.5 for $n+1$ variables implies it for $n$ variables.*

*Proof.* Add a variable $Y_{n+1}$ with $P_{n+1} = T$ and $H_{n+1,j} = 0$. The new residual algebra maps to $F(\mathcal{D})$ by $Y_{n+1} \mapsto 0$; transport a certificate and forget $Q_{n+1}$. □

So it suffices to treat $n = m \geq 2$, which gives at least three homogeneous coordinates below. For the rest of the proof fix a CFN ring $A$, residual data $\mathcal{D}$ in $m \geq 2$ variables, $F = F(\mathcal{D})$, and $I = (b_1, \ldots, b_t) \subseteq A$.

## 4. The projective setting

4.1. **Homogeneous equations.** Use homogeneous coordinates $X_1, \ldots, X_m, Z$ and $S = A[X_1, \ldots, X_m, Z]$. Let $d_i = \deg P_i$, choose $\delta_i$ larger than the degrees of all $H_{ij}$, and let $e_i = d_i + \delta_i$. Put

$$M_i = Z^{d_i} P_i(X_i/Z) = X_i^{d_i} + Z\bar{Q}_i, \qquad \mathbf{P}_i = Z^{\delta_i} M_i - \sum_j b_j \, Z^{e_i} H_{ij}(X/Z),$$

homogeneous of degree $e_i$, and $J = (\mathbf{P}_1, \ldots, \mathbf{P}_m)$. Then $\mathrm{deh}(\mathbf{P}_i) = 0$ in $F$: these are exactly the residual equations. We do not saturate $J$; the projective scheme it defines may have extra components at infinity.

4.2. **Čech cochains.** For a nonempty set $\sigma$ of coordinates let $S_\sigma$ be the localisation of $S$ at the product of the coordinates in $\sigma$, and $S_\sigma(k)$ its part of degree $k$. Then

$$C^p(k) = \prod_{|\sigma| = p+1} S_\sigma(k)$$

with the alternating differential $d$ form the Čech complex of $\mathcal{O}(k)$ on the standard charts. We represent an element of $S_\sigma(k)$ by its finitely supported Laurent coefficient function. This representation is faithful over every ring, including rings with zero divisors, because multiplication by a monomial only translates coefficients.

**Definition 4.1.** $JC^p(k)$ is the image of $\bigoplus_i C^p(k - e_i) \to C^p(k)$, $(y_i) \mapsto \sum_i \mathbf{P}_i y_i$. $I^r C^p(k)$ is the set of finite sums $\sum b^\alpha \omega_\alpha$ with $|\alpha| = r$. We write $x \equiv y$ for $x - y \in JC^p(k)$.

These are images inside the complex. They are not tensor products, and no flatness is assumed or needed. Both are preserved by $d$.

The section $Z^{-1}$ appears through the cochain $z^{\mathrm{init}}$ with $z^{\mathrm{init}}_Z = Z^{-1}$. The main intermediate result is:

**Theorem 4.2** (Closed lift)**.** *There is $\zeta \in C^0(-1)$ with*

$$d\zeta \in JC^1(-1), \qquad \zeta_Z - Z^{-1} \in IS_Z(-1) + JS_Z(-1).$$

Sections 5 and 6 prove it. Section 7 shows that it implies Theorem 3.5.

## 5. Finite-fibre sections

**Proposition 5.1.** *For every $r \geq 0$ there is an explicit $z_r \in C^0(-1)$ with*

1. $(z_r)_Z = Z^{-1}$;
2. $dz_r \in I^{r+1}C^1(-1) + JC^1(-1)$;
3. $z_r - z_0 \in IC^0(-1) + JC^0(-1)$.

*The witnesses for these memberships are explicit as well.*

*Proof.* On the chart $X_i$ put $u = Z/X_i \in S_{X_i}(0)$. Dividing by $X_i^{e_i}$,

$$\mathbf{P}_i/X_i^{e_i} = G_i - \sum_j b_j h_{ij}, \qquad G_i = u^{\delta_i} H_i, \qquad H_i = M_i/X_i^{d_i} = 1 + uQ_i,$$

with $Q_i, h_{ij} \in S_{X_i}(0)$. Hence $G_i \equiv \sum_j b_j h_{ij}$, and

$$G_i^p \in I^p + J \qquad \text{for every } p. \tag{5.1}$$

Fix $r$, let $p = r+1$, write $H_i^p = 1 + uQ_{i,p}$, and choose $N \geq \delta_i p$ for all $i$. Define

$$t_i = (-Q_{i,p})^{N+1} u^N, \qquad (z_r)_{X_i} = t_i/X_i, \qquad (z_r)_Z = Z^{-1}.$$

Then $u\,t_i = (1 - H_i^p)^{N+1} = 1 - H_i^p a_i$ for a polynomial $a_i$ in $H_i^p$, so

$$Z\,(z_r)_{X_i} = 1 - a_i H_i^p. \tag{5.2}$$

*Edge $\{Z, X_k\}$.* By (5.2), $(z_r)_{X_k} - Z^{-1} = -a_k H_k^p/Z$. On $S_{ZX_k}$ the element $u_k$ is invertible and $H_k^p = u_k^{-\delta_k p} G_k^p \in I^p + J$ by (5.1).

*Edge $\{X_k, X_l\}$.* Write $z_k = (z_r)_{X_k}$. Since $u_l t_l + a_l H_l^p = 1$,

$$z_k = z_k u_l t_l + z_k a_l H_l^p, \qquad z_l = z_l u_k t_k + z_l a_k H_k^p,$$

and $z_k u_l t_l = t_k t_l Z/(X_k X_l) = z_l u_k t_k$. Hence

$$z_k - z_l = z_k a_l H_l^p - z_l a_k H_k^p.$$

In the first term $t_k$ contains $u_k^N = (X_l/X_k)^N u_l^N$, and $u_l^N H_l^p = u_l^{N-\delta_l p} G_l^p \in I^p + J$. The second term is symmetric. This proves (2) with $p = r+1$.

*Compatibility.* On $Z$ the difference $z_r - z_0$ vanishes. On $X_k$ apply the same cross-term identity to $z_r$ and $z_0$: the products $z_{r,k}\,u\,t_{0,k}$ and $z_{0,k}\,u\,t_{r,k}$ agree, and each remaining term contains a factor $u^N H_k$ with $N \geq \delta_k$, which lies in $I + J$ by (5.1) with $p = 1$. □

No idempotent, no comparison between different bounds $N$, and no hypothesis on $A$ is used. Classically, $t_i/X_i$ is $Z^{-1}$ on the part of the chart where the fibre lies and 0 elsewhere; the formulas make this precise modulo $I^{r+1}$ without forming the decomposition.

## 6. Finite lifting

Here the CFN hypothesis on $A$ is used. Classically this step expresses finiteness of $\bigoplus_r H^1(W, I^r\mathcal{O}_W(-1))$ over the Rees algebra. We need only one relation among the classes of the $dz_r$.

6.1. **The Rees presentation.** Let $D = A[U_1, \dots, U_t]$, graded by $U$-degree, and $\mathrm{ev} : D \to A$, $U_j \mapsto b_j$. Elements of degree $k$ evaluate into $I^k$. Consider

$$\rho : D[X, Z] \longrightarrow (S/J)[T], \qquad U_j \longmapsto b_j T.$$

Its kernel $K$ is bihomogeneous, for the $(X, Z)$-degree and the $U$-degree. A polynomial $F$ of $U$-degree $k$ satisfies $\rho(F) = F(b, X, Z)T^k$, so

$$F \in K \iff F(b, X, Z) \in J. \tag{6.1}$$

**Lemma 6.1.** *$K$ is generated by a finite list $g_1, \dots, g_a$ of bihomogeneous polynomials, of degrees $\delta(g_i)$ in $(X, Z)$ and $f_i$ in $U$, with explicit identities $g_i(b, X, Z) = \sum_q h_{iq}\mathbf{P}_q$.*

*Proof.* $K = (J, U_1 - b_1T, \dots, U_t - b_tT) \cap A[U, X, Z]$ is an elimination ideal. Since $A[U, X, Z]$ is CFN, eliminating the last variable $T$ gives finitely many generators with membership witnesses. Replace each generator by its bihomogeneous components, which lie in $K$ because $K$ is bihomogeneous. □

We do not assume that $J$ and the relations $b_iU_j = b_jU_i$ generate $K$: that would require flatness.

**Lemma 6.2** (Localised kernel membership)**.** *Let $c \in C_D^p(k)$ be homogeneous of $U$-degree $k'$ with $\mathrm{ev}(c) \in JC_A^p(k)$. Then merely $c = \sum_i g_i u_i$ with $u_i \in C_D^p(k - \delta(g_i))$ of $U$-degree $k' - f_i$.*

*Proof.* Work on each of the finitely many charts $\sigma$. Clear one monomial denominator $X^L$ common to $c_\sigma$ and to the $J$-multipliers of $\mathrm{ev}(c_\sigma)$. The polynomial $X^Lc_\sigma$ has $U$-degree $k'$ and lies in $J$ after evaluation, so it is in $K$ by (6.1). Write it as $\sum g_iF_i$, take bihomogeneous components of the $F_i$, and divide by $X^L$. A cochain is a family indexed by charts, so these chart-wise lifts form a cochain; only finitely many choices are made. □

6.2. **Lifting the sections to the Rees algebra.** By Proposition 5.1(2), $dz_r = w_r + j_r$ with $j_r \in JC^1$ and $w_r = \sum_{|\alpha|=r+1} b^\alpha\omega_\alpha$. Put

$$c_r = \sum_{|\alpha|=r+1} U^\alpha\omega_\alpha \in C_D^1(-1),$$

of $U$-degree $r + 1$, with $\mathrm{ev}(c_r) = w_r$. Then $\mathrm{ev}(dc_r) = dw_r = -dj_r \in JC^2$, and Lemma 6.2 gives, merely,

$$dc_r = \sum_i g_i u_{r,i}. \tag{6.2}$$

6.3. **Noetherian dependence.**

**Theorem 6.3.** *Let $D$ be CFN, let $S_D = D[X_0, \dots, X_{\nu-1}]$ have $\nu \geq 3$ coordinates, and let $g_1, \dots, g_a$ be homogeneous of degrees $\delta_i \geq 0$. Let $c_0, c_1, \dots \in C_D^1(-1)$ each merely satisfy $dc_r = \sum_i g_i u_{r,i}$. Then merely there are $N$ and $q_0, \dots, q_{N-1} \in D$ with*

$$c_N - \sum_{i<N} q_i c_i \in d\,C_D^0(-1) + \sum_i g_i\, C_D^1(-1 - \delta_i).$$

The proof has four ingredients.

*(a) Contraction of free modules.* For any ring $D$ and any twist $k$, the complex $C_D^\bullet(k)$ is exact in degrees $0 < p < \nu - 1$, and in the top degree $\nu - 1$ a cochain is a coboundary when its coefficients at the all-negative monomials vanish. The contraction is monomial by monomial: a Laurent monomial whose set of negative coordinates is $\tau$ occurs only on charts $\sigma \supseteq \tau$. These charts form a cone, hence a contractible complex, unless $\tau$ is empty or is everything. In a fixed degree there are only finitely many all-negative monomials.

*(b) A bounded homogeneous resolution.* Coherence of $S_D$ gives homogeneous matrices

$$F_{\nu-2} \xrightarrow{A_{\nu-3}} \cdots \xrightarrow{A_1} F_1 \xrightarrow{A_0=(g_1,\ldots,g_a)} F_0 = S_D(-1)$$

between finite sums of shifted free modules, each column list generating the homogeneous kernel of the previous matrix. This is exact at $F_1, \ldots, F_{\nu-3}$; nothing is claimed at the left end, and no termination of a resolution is used. Exactness persists on every chart: clear a common denominator, apply kernel generation, and translate back.

*(c) Raising and reflection.* Given $dc = A_0 u_1$ with $u_1 \in C^2(F_1)$, we have $A_0\, du_1 = ddc = 0$, so chart-wise exactness gives $du_1 = A_1 u_2$. Continuing, we obtain $u_{\nu-2} \in C^{\nu-1}(F_{\nu-2})$, in top degree. Let $\mathrm{top}(u) \in D^v$ be its finitely many coefficients at all-negative monomials. Define $\Gamma(c, p)$ to mean that such a chain merely exists with $\mathrm{top}(u_{\nu-2}) = p$. Then $\Gamma$ is a $D$-submodule of $C^1 \times D^v$, and every $c_r$ has a partner. Moreover

$$\Gamma(c, 0) \implies c \in dC^0 + A_0 C^1(F_1).$$

Indeed, if $\mathrm{top}(u_{\nu-2}) = 0$, then (a) gives $u_{\nu-2} = dh_{\nu-2}$. Then $u_{\nu-3} - A_{\nu-3}h_{\nu-2}$ is closed below the top degree, hence equal to some $dh_{\nu-3}$. Descending this way, we reach $u_1 = A_1 h_2 + dh_1$. Then $c - A_0 h_1$ is closed in degree 1; since $1 < \nu - 1$, it equals some $dh_0$. This is where $\nu \geq 3$ is used.

*(d) Dependence with partners under truncation.* Let $D$ be CFN and $\Gamma \subseteq Z \times D^v$ a submodule such that $\Gamma(z, 0)$ implies $z \in B$, for a submodule $B$. If every $b_r$ merely has a partner, then merely $b_N - \sum_{i<N} q_i b_i \in B$ for some $N$ and $q_i$. The proof is by induction on free lists in $D^v$, from constructive noetherianity. At each step one mere partner is obtained inside the propositional goal. If it lies in the span of the earlier partners (decidable by strong discreteness), we have the relation; otherwise the free list grows. No sequence of partners is ever chosen.

Theorem 6.3 follows from (c) and (d), with $Z = C_D^1(-1)$. It has no finite-matrix-kernel or cohomology-generator computation: dependence is found directly in the ambient free module $D^v$.

**Corollary 6.4** (Weighted and evaluated form)**.** *For the $c_r$ of* (6.2) *there are $N$ and $r_i \in I^{N-i}$ $(i < N)$, $v \in I^{N+1}C^0(-1)$ and*

$$w_N - \sum_{i<N} r_i w_i \in dv + JC^1(-1).$$

*Proof.* Apply Theorem 6.3 over $D = A[U]$, which is CFN, with $\nu = m + 1 \geq 3$. Since $c_r$ has $U$-degree $r + 1$ and $g_i$ has $U$-degree $f_i$, project the relation to $U$-degree $N + 1$. Projection commutes with $d$ and with multiplication by $U$-homogeneous polynomials, so $q_i$ may be replaced by its part of degree $N - i$, the primitive by its part of degree $N + 1$, and the multipliers of $g_i$ by their parts of degree $N + 1 - f_i$. Now evaluate at $U = b$. Here $\mathrm{ev}(g_i) = \sum_q h_{iq}\mathbf{P}_q$, so the $g$-terms land in $JC^1$. Negative degrees give zero parts. □

6.4. **Correction.**

*Proof of Theorem 4.2.* Let $\zeta = z_N - \sum_{i<N} r_i z_i - v$. Then

$$d\zeta \equiv w_N - \sum_{i<N} r_i w_i - dv \equiv 0$$

modulo $JC^1(-1)$. Also $\zeta - z_0 = (z_N - z_0) - \sum r_i z_i - v$ lies in $IC^0 + JC^0$, because $N - i \geq 1$. On the chart $Z$ this gives $\zeta_Z - Z^{-1} \in IS_Z + JS_Z$, since $(z_0)_Z = Z^{-1}$. □

## 7. Extraction

**Proposition 7.1.** *A cochain $\zeta$ as in Theorem 4.2 gives degree-lowering data (DL) for $(F, y, b)$.*

*Proof.* Take one finite bound $\beta$ for the negative exponents occurring in the supports of $\zeta$, of its $J$-multipliers on the edges $\{Z, X_i\}$, and of the $I$- and $J$-multipliers in $\zeta_Z - Z^{-1}$. Put $k_Z = \beta_Z + 1$ and $k_i = \beta_i + 1$. Then

$$\zeta_Z = \sigma / Z^{k_Z}, \qquad (\zeta)_{X_i} = q_i / X_i^{k_i},$$

with $\sigma$ and $q_i$ polynomials, homogeneous of degrees $\beta_Z$ and $\beta_i$. Put $s = \deh(\sigma)$.

On the edge $\{Z, X_i\}$ the condition $d\zeta \in JC^1(-1)$ reads $q_i/X_i^{k_i} - \sigma/Z^{k_Z} = \sum_q \mathbf{P}_q m_q$. Multiplying by $Z^{k_Z} X_i^{k_i}$ makes every term polynomial, by the choice of $\beta$:

$$Z^{k_Z} q_i - X_i^{k_i} \sigma = \sum_q \mathbf{P}_q \, (Z^{k_Z} X_i^{k_i} m_q).$$

Dehomogenising kills the right side and gives $s\, y_i^{\beta_i + 1} = \deh(q_i)$, which is (DL) with $a_i = \beta_i$. Similarly, multiplying $\zeta_Z - Z^{-1} = \sum_j b_j v_j + \sum_q \mathbf{P}_q n_q$ by $Z^{k_Z}$ and dehomogenising gives $s = 1 + \sum_j b_j \deh(v_j')$. □

Only the overlaps of the chart $Z$ with the other charts are used here. All identities are proved as equalities of Laurent coefficients and then evaluated, which is valid over every ring.

## 8. Proof of Theorem 1.1

For CFN $A$ and $m \geq 2$, Proposition 5.1, Corollary 6.4 and the correction give Theorem 4.2. Proposition 7.1 turns this into the data (DL), and Proposition 2.4 gives a certificate, proving Theorem 3.5 for $m \geq 2$. Lemma 3.6 covers $m = 0, 1$. Apply this to $A_0 = \mathbb{Z}[C_1, \ldots, C_M]$ and transport to $E$ (Section 3); Lemma 2.2 concludes. □

All choices in the proof are finite: lifts of finitely many generators and coefficients, finitely many chart-wise lifts, finite supports, and one mere partner at a time inside the noetherian induction. The sequence $(z_r)$ is given by an explicit formula, so no countable choice is needed. Excluded middle is not used; the only decisions are on natural numbers, exponents, and membership in spans in $D^v$ (strong discreteness).

## 9. The formalisation

The proof is checked in Unit, the proof checker of `https://github.com/coquand/unit`, a dependent type theory with segments and hidden arguments. The design is by Coquand and the code was written by Claude. All files are in `unit/sag/`.

9.1. **Foundations.** The development is parametrised by the segment `Theory` of `sag/`: univalence, replacement (images) and surjections. Function extensionality and propositional univalence are derived from these. No new axiom is introduced. There is no choice principle: finite choice (`FinCat.fchoice`) is proved by induction. There is no excluded middle, countable choice, synthetic algebraic geometry axiom or resizing. "Merely" is propositional truncation (`Truncation.Trunc`), and rings are set-rings. Equality in $R$ is never decided.

9.2. **The main theorem.** The final file is `zmt-general/FinalZMTGeneral.seg`:

```
module Theorem (R : CRing) (setR : isSet R.R)
 (I : IdealPred[R]) (propI : (r : R.R) -> isProp (I.In r))
 (Ea : Algebra over R) (n : Nat) (x : Fin n -> Ea.ring.R)
 (gen : (e : Ea.ring.R) -> || (H : Expr (Fin n)) * Id (eval x H) e ||)
 (fo : || QuotientFibre.Over[R,setR,I,propI,Ea].Finite ||) {
 Conclusion = (s : E) * (InIE (s - 1) *
     (ModuleFinite[R, B_s] * (InB_s s * isEquiv (phi : B_s[1/s] -> E[1/s]))))
 result : || Conclusion || }
```

(abbreviated). Here $B_s = R[s, sx_1, \ldots, sx_n]$ is the image subalgebra, with carrier a subtype of $E$. The map $\varphi$ is the canonical ring map of localisations, and the conclusion states that this particular map is an equivalence. Finiteness of $E/IE$ is `ModuleFinite` over the quotient rings $R/I$ and $E/IE$ of `cffr/QuotientRing`. The same file proves

```
universal : (n : Nat) -> CFNDescent.UniversalZMT n
```

which is Theorem 3.5 at $A_0 = \mathbb{Z}[C_1, \ldots, C_M]$: for all residual data $\mathcal{D}_0$, merely a `ZMTCert`. Earlier milestones had this principle as an explicit parameter; it is now proved, and `result` has no parameter beyond those displayed.

9.3. **Correspondence with the proof.** Each directory is a separate checked development, linked to its predecessors by relative symbolic links; earlier files are unchanged.

| Mathematics | Unit (directory / module) |
|---|---|
| Lemma 2.2 | `zmt-descent`: `CertificateSubalgebra`, `CertificateLocalisation`, `FiniteFibreLocalisation` |
| Lemma 2.3 | `exterior/DeterminantTrick`, `zmt-descent/CharacteristicMonic` |
| Proposition 2.4 | `zmt-certificate-extract-general`: `GenericDegreeBounds` (dichotomy), `GenericFilteredSpan` (the module $V$), `GenericDegreeLoweringCertificate` |
| Lemma 3.2, $E/IE$ finite | `zmt-descent`: `FiniteFibreCertificates.residual`, `QuotientFibre.bridge` |
| Universal data, transport | `zmt-descent`: `ResidualAlgebra`, `UniversalResidualAlgebra`, `CoefficientBase` ($\mathbb{Z}[C]$ is CFN), `CertificateTransport`, `CFNDescent` |
| Lemma 3.6 | `zmt4-completion/ResidualPadding`, `zmt-general/UniversalArityGlue` |
| Homogeneous equations | `zmt-sections-general`: `HomogeneousTermsGeneral`, `GenericMonicSetup` |
| Chart rings, Čech cochains | `zmt-laurent-general` (actual localisations of the polynomial tower, Laurent coefficients, restrictions); `zmt-cech-general` (`GenericLaurent`, `GenericCube`, `GenericCech`) |
| $JC^p$, $I^rC^p$ | `zmt-rees-general`: `ImageQuotient`, `GenericReesChart`; `FiniteIdealPowers` witness trees |
| Proposition 5.1 | `zmt-sections-general`: `GenericMonicChartRelation` (5.1), `GenericSectionData`, `GenericSectionErrors`, `GenericSectionAssembly`, `GenericFiniteFibreSections`; `zmt-assembly-general` |
| Rees kernel list | `zmt-rees-general`: `GenericReesRings`, `GenericReesList`, `GenericReesExchange` (reusing `ReesFiniteList4` and `DegreeCor`) |

| Mathematics | Unit (directory / module) |
|---|---|
| Lemma 6.2, lifts $c_r$ | `GenericLocalizedReesMembership`, `GenericLocalizedReesLift`, `GenericReesCochainMembership`, `GenericReesPowerLift` |
| Theorem 6.3(a) | `zmt-cech-general` (contraction $dh + hd + \pi = \mathrm{id}$) |
| Theorem 6.3(b) | `zmt-syzygies-general` (`HomogeneousKernel`), `zmt-certificate-general` (`FinitePolynomialResolution`, `LocalizedSyzygies`) |
| Theorem 6.3(c) | `zmt-certificate-general`: `CertificateExtension`, `FiniteZigzagCertificate`, `CechZigzagSchedule`, `ScalarResolutionDependence` |
| Theorem 6.3(d) | `zmt-lifting/CertifiedDependence` |
| Corollary 6.4 | `zmt-weighted-general`: `GenericWeightedDependence`, `EvaluatedWeightedDependence` |
| Correction, Theorem 4.2 | `zmt-rees-general`: `GenericReesFactor`, `GenericReesCorrection`, `GenericReesFromSections`; `zmt-correction-general` |
| Proposition 7.1 | `zmt-extract-exponents-general` (`ExtractionBounds`), `zmt-extract-general` (`PrescribedPolynomialization`, `GenericClearedExtraction`), `zmt-general/GenericSectionExtraction` |
| Assembly | `zmt-general/FinalZMTGeneral` |

9.4. **Representation choices.** Some formal choices differ from the text, without changing the mathematics.

- Exponents are functions $\mathrm{Fin}\, n \to \mathbb{N}$ or $\mathrm{Fin}\, n \to \mathbb{Z}$. Subsets of coordinates are structural bitsets. The newest coordinate is $Z$ and comes first in the orientation, so $d(a, b) = (da, a - db)$ uniformly in $n$, with no sign tables.
- Chart elements are finitely supported Laurent coefficient functions with a support condition. The native localisation of the polynomial tower is compared with them by coefficient extensionality.
- Ideal powers $I^r$ are explicit finite witness trees; $J$-membership is an image of the action of the original $\mathbf{P}_q$. There are no quotient modules and no tensor products.
- The polynomial ring is an iterated tower $A[X_1] \cdots [X_n]$. Homogeneous elements are handled through finite term lists, and deh is evaluation of the tower. Monic polynomials are data with a literal leading coefficient 1.
- The dependence theorem is proved for $\nu = \mathit{extra} + 3$ coordinates and the sections for $m + 1$ coordinates; one transport along $\mathit{extra}+3 = (\mathit{extra}+2)+1$ (`GenericCorrectionTransport`) identifies the two interfaces.

9.5. **Validation.** On 30 October 2026, a fresh run of the checker on `zmt-general/All.seg`, which loads every development above, checked 18 779 declarations with 0 failures in 13.1 s. The checking server reported 359 modules, 0 failed. The opening audit (`-opens`) confirms, for every declaration owned by the final directory, that it unfolds only the listed definitions. Each owned declaration opens at most 35 definitions and takes at most 0.02 s. The intermediate directories carry their own audits; for example `zmt-rees-general` owns 1121 declarations and `zmt-certificate-general` owns 1132. All commands were bounded to 55 s.

The regressions run the final theorem with no assumed principle:

- $R = E = \mathbb{Z}/4$, $I = (2)$, every generator equal to 2, for $n = 0, \ldots, 5$, so the base is not reduced;

- the zero fibre $I = R$, for $n = 0$ and $n = 4$;
- extraction with no affine variables, and extraction from the constructed correction for arbitrary residual data in four variables;
- certificates over $\mathbb{Z}$ for $m = 0, 4$, and over $\mathbb{Z}/4$ with nonzero nilpotent generators.

Deliberate mutations are rejected. The final conclusion with $s + 1$ in place of $s - 1$ fails, and so does the zero map in place of the canonical localisation map. Earlier tracks reject a reversed edge orientation, a wrong homogeneous degree, the bound $d = D$ in place of $D + \sum a_i$, the correction $c + ap$ in place of $c - ap$, and a sign error in a coefficient degree shift.

9.6. **Scope.** The checked statement is Theorem 1.1 for every finite number of generators. Its hypotheses are an ideal with propositional membership and the quotient-ring model of $E/IE$ in `cffr`. A caller with a different quotient construction needs an isomorphism to this one. The inverse of $\varphi$ is not packaged separately as a ring map; it is one, since $\varphi$ is a bijective ring map. An earlier, independent development (`zmt4-completion`) proves the case of at most three generators with four fixed homogeneous coordinates; the general proof uses only its dimension-independent helpers, such as the padding lemma.

## References

[1] M. Alonso, T. Coquand and H. Lombardi, *Revisiting Zariski Main Theorem from a constructive point of view*, J. Algebra 406 (2014), 46–68.
[2] T. Coquand and H. Persson, *Gröbner bases in type theory*, Types for Proofs and Programs, LNCS 1657 (1999), 33–46.
[3] T. Coquand, *Local structure of étale algebras*, manuscript (`zmt2-revised.pdf`).
[4] The Stacks Project, Section 30.20, *The theorem on formal functions*.
[5] Internal notes in `unit/doc/`: `zmt-formal-cohomology.tex` (the plan of this proof) and the `zmt-*-handoff.txt` files.